

Vehicle Damage Description and NLP Entity Annotation Dataset

AI-Based Vehicle Damage Assessment and Insurance Assistant

1. Introduction

This document defines the damage-description dataset used for the Natural Language Processing (NLP) component of the **AI-Based Vehicle Damage Assessment and Insurance Assistant**.

The dataset is designed to generate realistic descriptions of visible vehicle damage corresponding to the six damage categories used in the CarDD vehicle damage dataset:

1. Dent
2. Scratch
3. Crack
4. Glass Shatter
5. Lamp Broken
6. Tire Flat

Each description is associated with domain-specific NLP entities such as **DAMAGE_TYPE**, **DAMAGE_SEVERITY**, and **DAMAGED_PART**. These annotations will be used to train a custom Named Entity Recognition (NER) model capable of extracting structured damage information from user-provided accident descriptions.

2. Dataset Objectives

The primary objectives of this dataset are:

- To represent common vehicle damage using natural-language descriptions.
 - To provide training data for domain-specific NLP entity extraction.
 - To identify the type of damage reported by the user.
 - To identify the affected vehicle component when explicitly mentioned.
 - To identify severity-related terms such as *minor*, *small*, *deep*, *large*, and *severe*.
 - To convert unstructured accident descriptions into structured information.
 - To support comparison between user-reported damage and computer-vision-based damage detection.
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3. Entity Categories

The following entity categories are used in the NLP annotation scheme.

Entity	Description
DAMAGE_TYPE	Describes the type or nature of the vehicle damage.
DAMAGE_SEVERITY	Describes the apparent severity or extent of the damage.
DAMAGED_PART	Identifies the vehicle component or area affected by the damage.

Examples

Text	Entity
dent	DAMAGE_TYPE
scratches	DAMAGE_TYPE
cracked	DAMAGE_TYPE
broken	DAMAGE_TYPE
flat	DAMAGE_TYPE
small	DAMAGE_SEVERITY
minor	DAMAGE_SEVERITY
deep	DAMAGE_SEVERITY
large	DAMAGE_SEVERITY
severely	DAMAGE_SEVERITY
body panel	DAMAGED_PART
window	DAMAGED_PART
lamp	DAMAGED_PART
tire	DAMAGED_PART

4. Damage Category: Dent

Description

A dent refers to a visible indentation or deformation of a vehicle's exterior surface, commonly caused by an impact or collision.

Sample Damage Descriptions

1. The vehicle has a visible dent.
2. There is a dent on the car body.
3. A noticeable dent can be seen on the damaged panel.
4. The body panel has a deep dent.
5. A small dent is visible on the vehicle.
6. The car has a large deformation caused by an impact.
7. There is an indentation on the vehicle body.
8. The damaged area shows a significant dent.

Entity Mapping

Term	Entity
dent	DAMAGE_TYPE
small	DAMAGE_SEVERITY
large	DAMAGE_SEVERITY
deep	DAMAGE_SEVERITY
body panel	DAMAGED_PART
vehicle body	DAMAGED_PART
damaged panel	DAMAGED_PART

5. Damage Category: Scratch

Description

A scratch refers to a visible mark, abrasion, or removal of surface paint or material from the exterior of a vehicle.

Sample Damage Descriptions

1. The vehicle has visible scratches on its body.
2. Several scratches can be seen on the damaged area.
3. There is a long scratch across the vehicle panel.
4. The car body has multiple surface scratches.
5. A visible scratch is present on the exterior.
6. The damaged panel shows several thin scratches.
7. There are minor scratches on the vehicle body.
8. The exterior paint appears scratched.

Entity Mapping

Term	Entity
scratch	DAMAGE_TYPE
scratches	DAMAGE_TYPE
scratched	DAMAGE_TYPE
minor	DAMAGE_SEVERITY
long	DAMAGE_SEVERITY
multiple	DAMAGE_SEVERITY
body	DAMAGED_PART
panel	DAMAGED_PART
damaged area	DAMAGED_PART
exterior	DAMAGED_PART

6. Damage Category: Crack

Description

A crack refers to a visible fracture or split in a vehicle component or surface.

Sample Damage Descriptions

1. A crack is visible on the vehicle.
2. The damaged area has a visible crack.
3. There is a long crack across the surface.
4. A deep crack can be seen in the damaged area.
5. The vehicle component shows a visible fracture.
6. A narrow crack extends across the damaged region.
7. The exterior component has developed a crack.

Entity Mapping

Term	Entity
crack	DAMAGE_TYPE
fracture	DAMAGE_TYPE
deep	DAMAGE_SEVERITY
long	DAMAGE_SEVERITY
narrow	DAMAGE_SEVERITY
damaged area	DAMAGED_PART
damaged region	DAMAGED_PART
vehicle component	DAMAGED_PART
exterior component	DAMAGED_PART

7. Damage Category: Glass Shatter

Description

Glass shatter represents significant damage to vehicle glass, including shattered sections, broken glass, and extensive cracks.

Sample Damage Descriptions

1. The vehicle glass is shattered.
2. The glass has extensive cracks and shattered sections.
3. The window appears severely shattered.
4. Several broken pieces are visible in the glass.
5. The vehicle window has significant glass damage.

6. The glass surface is heavily damaged and shattered.
7. The window has multiple cracks and broken sections.

Entity Mapping

Term	Entity
glass damage	DAMAGE_TYPE
shattered	DAMAGE_TYPE
broken	DAMAGE_TYPE
cracks	DAMAGE_TYPE
severely	DAMAGE_SEVERITY
heavily	DAMAGE_SEVERITY
extensive	DAMAGE_SEVERITY
significant	DAMAGE_SEVERITY
window	DAMAGED_PART
glass	DAMAGED_PART

8. Damage Category: Lamp Broken

Description

Lamp broken refers to visible damage to a vehicle's lighting components, including broken, cracked, or severely damaged lamp assemblies.

Sample Damage Descriptions

1. The vehicle lamp is broken.
2. The headlamp has visible damage.
3. The lamp assembly is cracked and broken.
4. The vehicle light has been severely damaged.
5. A broken lamp is visible on the vehicle.
6. The light assembly has sustained damage.
7. The lamp cover appears broken.
8. The vehicle's lighting component is damaged.

Entity Mapping

Term	Entity
lamp	DAMAGED_PART
headlamp	DAMAGED_PART
light	DAMAGED_PART
lighting component	DAMAGED_PART
broken	DAMAGE_TYPE

Term	Entity
damaged	DAMAGE_TYPE
cracked	DAMAGE_TYPE
severely	DAMAGE_SEVERITY

9. Damage Category: Tire Flat

Description

Tire flat refers to a visibly deflated or flattened vehicle tire, potentially resulting from a puncture, loss of air pressure, or other damage.

Sample Damage Descriptions

1. The vehicle has a flat tire.
2. One of the tires appears deflated.
3. The tire is visibly flat.
4. The vehicle's tire has lost its normal shape.
5. A severely deflated tire is visible.
6. The tire appears damaged and flat.
7. One tire is completely deflated.

Entity Mapping

Term	Entity
tire	DAMAGED_PART
flat tire	DAMAGED_PART
flat	DAMAGE_TYPE
deflated	DAMAGE_TYPE
damaged	DAMAGE_TYPE
severely	DAMAGE_SEVERITY
completely	DAMAGE_SEVERITY

10. Consolidated Damage Taxonomy

Damage Category	DAMAGE_TYPE	DAMAGE_SEVERITY	DAMAGED_PART
Dent	dent, indentation, deformation	small, large, deep, significant	body, body panel, damaged panel
Scratch	scratch, scratches, scratched	minor, long, multiple, thin	body, panel, exterior

Damage Category	DAMAGE_TYPE	DAMAGE_SEVERITY	DAMAGED_PART
Crack	crack, fracture	deep, long, narrow	component, damaged area, exterior
Glass Shatter	shattered, broken, glass damage, cracks	severe, heavy, extensive, significant	glass, window
Lamp Broken	broken, damaged, cracked	severe	lamp, headlamp, light assembly
Tire Flat	flat, deflated, damaged	severe, completely	tire

11. NLP Processing Objective

The descriptions in this dataset will be used to train a domain-specific Named Entity Recognition model.

For example, given the input:

"The rear body panel has a deep dent."

The NLP model should identify:

```
rear body panel → DAMAGED_PART
deep           → DAMAGE_SEVERITY
dent           → DAMAGE_TYPE
```

Similarly:

"The vehicle has several scratches on the exterior."

should produce:

```
several        → DAMAGE_SEVERITY
scratches      → DAMAGE_TYPE
exterior       → DAMAGED_PART
```

The extracted information can then be converted into structured JSON for downstream processing.

12. Example Structured Output

Input

"The vehicle has a deep dent on the damaged body panel."

NLP Output

```
{  
  "damage_type": "DENT",  
  "damage_severity": "DEEP",  
  "damaged_part": "BODY_PANEL"  
}
```

Input

"The window appears severely shattered."

NLP Output

```
{  
  "damage_type": "GLASS_SHATTER",  
  "damage_severity": "SEVERE",  
  "damaged_part": "WINDOW"  
}
```

Input

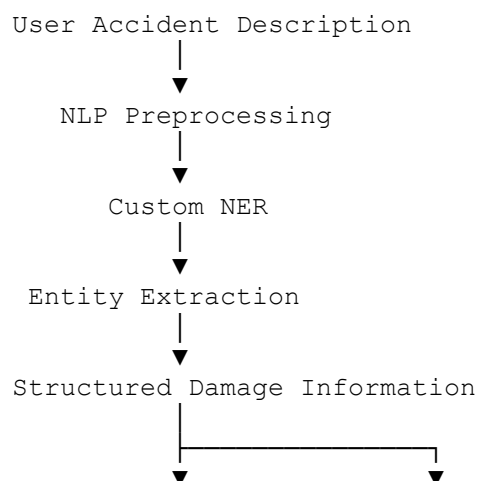
"The vehicle's left lamp is broken."

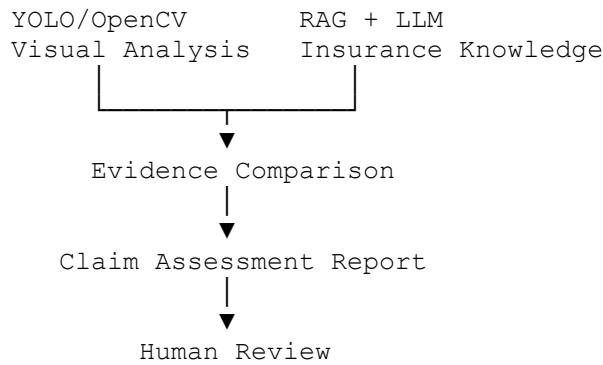
NLP Output

```
{  
  "damage_type": "BROKEN",  
  "damage_severity": null,  
  "damaged_part": "LAMP"  
}
```

13. Role in the Overall AI System

The NLP module will form one component of the complete AI-based vehicle damage assessment system.





The NLP output can therefore be compared with computer-vision predictions to identify potential inconsistencies between the user's description and the visible vehicle damage.

14. Dataset Development Recommendation

For model training, these initial descriptions should be expanded into a larger annotated dataset.

The recommended process is:

1. Use the CarDD damage categories as the initial damage taxonomy.
 2. Generate multiple natural-language descriptions for each damage category.
 3. Manually review generated descriptions.
 4. Annotate entities using `DAMAGE_TYPE`, `DAMAGE_SEVERITY`, and `DAMAGED_PART`.
 5. Split the dataset into training, validation, and test sets.
 6. Train a domain-specific NER model.
 7. Evaluate the model using precision, recall, and F1-score.
 8. Test the model on previously unseen accident descriptions.
 9. Integrate the NLP output with the computer-vision damage detection module.
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15. Important Annotation Principle

Entity annotations should only represent information explicitly stated or reliably supported by the data.

For example:

"A dent is visible on the damaged area."

should **not** automatically be annotated as:

rear bumper → `DAMAGED_PART`

unless the image or source annotation confirms that the damage is on the rear bumper.

This prevents the NLP dataset from introducing unsupported information and improves the reliability of the insurance-assistance system.

Conclusion

The proposed damage-description dataset provides a structured foundation for developing the NLP component of the AI-Based Vehicle Damage Assessment and Insurance Assistant. By mapping natural-language descriptions to standardized damage types, severity indicators, and affected vehicle components, the system can transform unstructured user descriptions into structured information suitable for downstream claim processing.

The combination of NLP-based user-reported evidence with computer-vision-based damage detection can enable the system to identify inconsistencies, generate evidence-grounded claim summaries, and support human claim reviewers while maintaining a human-in-the-loop decision process.